

| Question |     |      |    | Marking details                                                                                                                                                                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|-----|------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |    |                                                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 11       | (a) | (i)  |    | $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} \quad (1)$<br>$2\text{MnO}_4^- + 5\text{H}_2\text{C}_2\text{O}_4 + 6\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 10\text{CO}_2 + 8\text{H}_2\text{O} \quad (1)$                                                                                                                                                     | 1               | 1   |     | 2     |       |      |
|          |     | (ii) | I  | 31.67 cm <sup>3</sup><br>accept 31.65 cm <sup>3</sup>                                                                                                                                                                                                                                                                                                                                                        |                 | 1   |     | 1     | 1     | 1    |
|          |     |      | II | moles of manganate(VII) = $\frac{31.67 \times 0.0505}{1000} = 1.599 \times 10^{-3} \text{ mol} \quad (1)$<br>ECF possible from mean volume<br>moles of oxalic acid = $1.599 \times 10^{-3} \times 2.5 = 4.00 \times 10^{-3} \text{ mol} \quad (1)$<br>ECF possible from equation<br>concentration of saturated solution = $\frac{4.00 \times 10^{-3}}{0.025} \times 10 = 1.60 \text{ mol dm}^{-3} \quad (1)$ |                 | 3   |     | 3     | 2     |      |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks available |     |     |       |       |      |
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|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b) |  |  | <p><b>Indicative content</b></p> <ol style="list-style-type: none"><li>1. Oxalic acid has two acidic protons</li><li>2. The <math>K_a</math>/acidity/ease of removal of each proton is different so they give two different vertical regions</li><li>3. First vertical region occurs at half the volume of second as each occurs after removing same number of protons</li><li>4. Flat regions from 5-10 and 20-25 due to formation of buffer</li><li>5. Alice, Brychan and David methods will work</li><li>6. Indicators only work if they change colour within a vertical region</li><li>7. Alice's method allows the first and second equivalence points to be found by plotting a curve</li><li>8. Brychan's indicator will change colour fully during the second vertical region</li><li>9. David's indicator will change colour fully during the first vertical region</li><li>10. Carys's method will not work as indicator will change colour gradually as sodium hydroxide is added</li></ol> |                 |     |     |       |       |      |
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 | 3   | 3   | 6     |       | 3    |

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                   |   |   |   |    |   |   |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---|---|---|----|---|---|
|  |  |  | <p><b>5-6 marks</b><br/>The candidate includes at least six relevant points<br/><i>The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary is used accurately throughout.</i></p> <p><b>3-4 marks</b><br/>The candidate includes at least five relevant points<br/><i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i></p> <p><b>1-2 marks</b><br/>The candidate includes at least three relevant points<br/><i>The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i></p> <p><b>0 marks</b><br/><i>The candidate does not make any attempt or give an answer worthy of credit.</i></p> |                   |   |   |   |    |   |   |
|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Question 11 total | 1 | 8 | 3 | 12 | 3 | 4 |